

- a What was the most popular music type for girls?
- b What percentage of girls selected this type as their favourite music?
- c What was the most popular music type for boys?
- d What percentage of boys selected this type as their favourite?
- e Why is a bar graph more useful than a column graph for displaying this data?

- 6** A class of 20 students keeps a record of all the TV they watch in a week. The students combine their results and divide them into the categories shown at right.

- a Use a spreadsheet to draw a column graph of the data.
- b If there are a total of 200 hours of TV watched, how many hours do the students spend watching music shows?

| Category of show | Percentage of time |
|------------------|--------------------|
| Movies           | 30                 |
| Drama series     | 20                 |
| Sitcoms          | 15                 |
| Music shows      | 10                 |
| Documentaries    | 10                 |
| Current affairs  | 5                  |
| News             | 5                  |

LINKS TO  
Example 13

- 7** Students were surveyed on what they were doing at 7 pm last night. The results are summarised in the table below.

| Activity           | Frequency | Angle |
|--------------------|-----------|-------|
| Watching TV        | 4         |       |
| Eating dinner      | 2         |       |
| On the phone       | 5         |       |
| At the computer    | 5         |       |
| Reading            | 4         |       |
| Playing music      | 1         |       |
| Playing sport      | 3         |       |
| Listening to music | 2         |       |
| Doing homework     | 10        |       |
| <b>Total</b>       | <b>36</b> |       |

- a If this information was presented in a pie chart, how many degrees would represent one student?
- b In the right-hand column in the table above, fill in the angle each activity would make in a pie chart.
- c Draw a pie chart representing the information in the table above.

LINKS TO  
Example 13

- 8 Zeev kept a record of how he spent his time in a day. He wishes to show this data in a pie graph. Complete the table below showing the sector angle corresponding to each activity.

| Activity | Hours | Angle |
|----------|-------|-------|
| Sleeping | 9     |       |
| Eating   | 3     |       |
| In class | 5     |       |
| Sport    | 2     |       |
| TV       | 2     |       |
| Homework | 2     |       |
| Other    | 1     |       |

LINKS TO  
Example 14

- 9 A group of 36 Year 7 students were surveyed about the number of pets they have at home. The results of the survey are given below.

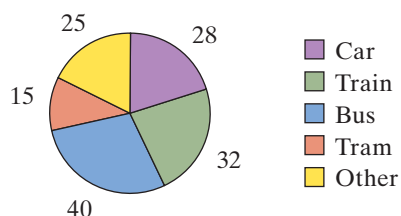
1, 1, 2, 2, 0, 4, 3, 2, 2, 0, 1, 1, 2, 3, 4, 5, 0, 2, 1, 3, 1, 2, 1, 5, 4, 2, 2, 2, 0, 0, 1, 1, 2, 1, 2, 4

- What is the variable here and what values can it take?
- Construct a frequency table for the data set.
- Construct a pie chart to represent the data.

- 10 The following pie chart shows how a group of Year 7 students travel to school.

- Do the numbers on each sector represent percentages or numbers of students? How do you know?
- How many students travel to school by car?
- What is the most common method of transport?
- What percentage of students take the train?

How students travel to school



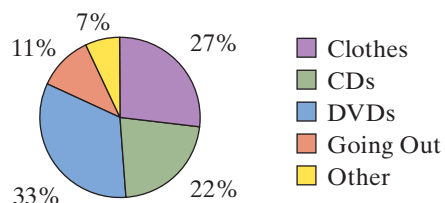
LINKS TO  
Example 15

- 11 The pie chart to the right shows how Daniel spent the money he received for his birthday.

If Daniel received \$450 for his birthday

- how much did he spend on clothes?
- how much did he spend on DVDs?
- What information about the DVDs he bought is not clear from this graph?

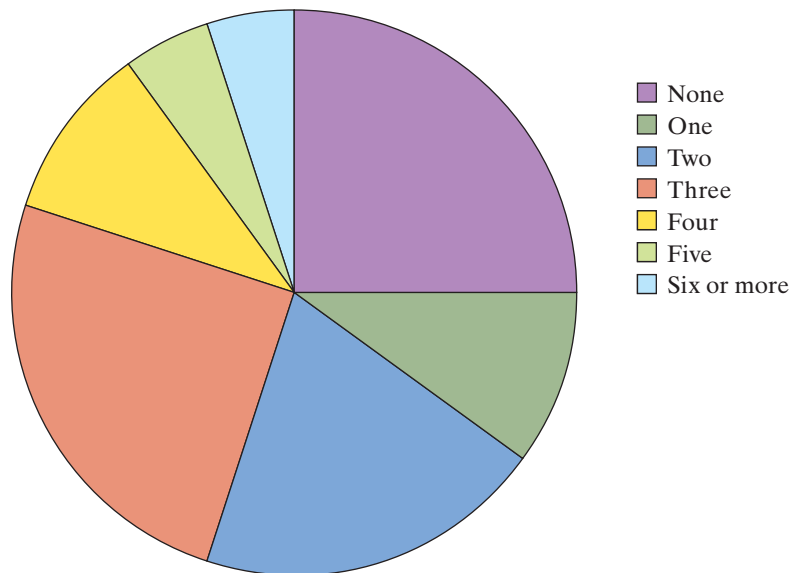
How Daniel spent his birthday money



- 12** Town planners surveyed a community to find out how many children lived in each household. The results are shown below.

**13.4**

**Number of children in households**



- a** Measure the angle for each sector and complete the table.  
**b** What percentage of households has no children?  
**c** What percentage of households has more than five children?

| Children | Angle | Percentage |
|----------|-------|------------|
| 0        |       |            |
| 1        |       |            |
| 2        |       |            |
| 3        |       |            |
| 4        |       |            |
| 5        |       |            |
| >5       |       |            |

## exercise 13.4

## challenge

- 13** This pie graph represents some data collected by your class. Design a question that the data on the pie graph might represent, and provide a frequency table of responses. Also add labels and a legend so that the graph makes sense.

